QUBO Formulation: N-Queens Exact-Cover Problem

N-Queens Exact-Cover Problem

Problem Statement. Given an $N \times N$ chessboard, place N queens such that no two queens attack each other. A queen at position (i, j) attacks all squares in row i, column j, and both diagonals passing through (i, j). The task is to select exactly N squares satisfying these non-attacking conditions.

Decision Variables. We introduce a binary variable $x_{i,j} \in \{0, 1\}$ for each square (i, j) on the board, where $i, j \in \{0, 1, \ldots, N-1\}$. The variable $x_{i,j} = 1$ if and only if a queen is placed at row i and column j; otherwise $x_{i,j} = 0$.

QUBO Derivation. The problem is a feasibility (exact-cover) problem with no intrinsic cost function. We formulate it as minimizing the total number of violated constraints by converting each constraint into a quadratic penalty term. Let $\lambda > 0$ be the penalty weight.

Step 1 – Objective. Since there is no original objective function, we minimize the sum of squared violations of all constraints:

$$\min_{x \in \{0,1\}^{N^2}} \left[\text{Row Violations} + \text{Column Violations} + \text{Diagonal Violations} + \text{Anti-Diagonal Violations}\right]. \tag{1}$$

Step 2 – Constraint Algebra. The constraints are expressed as follows:

$$\text{Row } i: \quad \sum_{j=0}^{N-1} x_{i,j} = 1, \quad \forall i \in \{0, \ldots, N-1\}, \tag{2}$$

$$\text{Column } j: \quad \sum_{i=0}^{N-1} x_{i,j} = 1, \quad \forall j \in \{0, \ldots, N-1\}, \tag{3}$$

$$\text{Main Diagonal } k: \quad \sum_{(i,j) \in D_k} x_{i,j} \leq 1, \quad \forall k \in \{-(N-1), \ldots, N-1\}, \tag{4}$$

$$\text{Anti-Diagonal } k: \quad \sum_{(i,j) \in AD_k} x_{i,j} \leq 1, \quad \forall k \in \{0, \ldots, 2N-2\}, \tag{5}$$

where $D_k = \{(i, j) : i - j = k\}$ and $AD_k = \{(i, j) : i + j = k\}$.

Step 3 – Penalty Expansion. We convert each constraint to a quadratic penalty. For equality constraints (rows and columns), we use the squared form $\lambda(\sum x - 1)^2$. For inequality constraints (diagonals), we penalize pairwise conflicts using $\lambda \sum_{u<v} x_u x_v$, which is zero when at most one variable is active and positive otherwise.

Row constraints:

$$P_{\text{row}}(i) = \lambda \left(\sum_{j=0}^{N-1} x_{i,j} - 1 \right)^2 \tag{6}$$

$$= \lambda \left(\sum_{j=0}^{N-1} x_{i,j}^2 - 2 \sum_{0 \le j < l \le N-1} x_{i,j} x_{i,l} + 1 \right) \tag{7}$$

$$= \lambda \sum_{j=0}^{N-1} x_{i,j} - 2\lambda \sum_{0 \le j < l \le N-1} x_{i,j} x_{i,l} + \lambda, \tag{8}$$

where we used the idempotent property $x_{i,j}^2 = x_{i,j}$.

Column constraints:

$$P_{\text{col}}(j) = \lambda \left(\sum_{i=0}^{N-1} x_{i,j} - 1 \right)^2 \tag{9}$$

$$= \lambda \sum_{i=0}^{N-1} x_{i,j} - 2\lambda \sum_{0 \le i < i' \le N-1} x_{i,j} x_{i',j} + \lambda. \tag{10}$$

Diagonal constraints:

$$P_{\text{diag}}(k) = \lambda \sum_{(i,j)<(i',j') \in D_k} x_{i,j} x_{i',j'}, \tag{11}$$

$$P_{\text{adiag}}(k) = \lambda \sum_{(i,j)<(i',j') \in AD_k} x_{i,j} x_{i',j'}. \tag{12}$$

Step 4 – Combined Hamiltonian. Summing all penalty contributions yields the QUBO Hamiltonian $H(x)$:

$$H(x) = \sum_{i=0}^{N-1} P_{\text{row}}(i) + \sum_{j=0}^{N-1} P_{\text{col}}(j) + \sum_{k} P_{\text{diag}}(k) + \sum_{k} P_{\text{adiag}}(k). \tag{13}$$

Step 5 – Q Matrix and Offset. Collecting linear and quadratic terms from the expanded penalties, we read off the QUBO coefficients $Q_{u,v}$ and constant offset c such that $H(x) = \sum_{u \le v} Q_{u,v} x_u x_v + c$. Identifying each board square (i, j) with a unique index u, we

obtain:

$$Q_{(i,j),(i,j)} = 2\lambda, \tag{14}$$
$$Q_{(i,j),(i,l)} = -2\lambda \quad \text{for } j \neq l, \tag{15}$$
$$Q_{(i,j),(i',j)} = -2\lambda \quad \text{for } i \neq i', \tag{16}$$
$$Q_{(i,j),(i',j')} = \lambda \quad \text{if } (i,j) \text{ and } (i',j') \text{ share a main or anti-diagonal}, \tag{17}$$
$$Q_{(i,j),(i',j')} = 0 \quad \text{otherwise}, \tag{18}$$
$$c = 2N\lambda. \tag{19}$$

Penalty Weight Justification. The penalty weight λ must be large enough to ensure that any energy reduction from satisfying a constraint outweighs the penalty incurred by violating another. Since the problem contains no intrinsic objective terms (it is purely a feasibility problem), there is no trade-off between objective improvement and constraint violation. Therefore, any $\lambda > 0$ strictly penalizes infeasible configurations relative to feasible ones. We set $\lambda = 1$ for simplicity, which matches the standard scaling for exact-cover QUBOs.

Correctness Argument. Minimizing $H(x)$ over $\{0,1\}^{N^2}$ recovers the optimal solution because: (i) any feasible placement satisfies all constraints exactly, yielding zero penalty terms and thus $H(x) = 0$; (ii) any infeasible placement violates at least one constraint, resulting in strictly positive penalty terms and thus $H(x) > 0$. Consequently, the global minima of $H(x)$ are exactly the feasible queen placements, and the minimum energy is 0.

Illustrative Example. Consider the concrete instance $N = 4$. The board has $4 \times 4 = 16$ squares, indexed 0 to 15 via $u = 4i + j$. The decision variables are $x_{0,0}, \ldots, x_{3,3} \in \{0,1\}$.

Using $\lambda = 1$, the general formulas instantiate as follows:

- **Row penalties:** For each $i \in \{0,1,2,3\}$, $P_{\text{row}}(i) = \sum_{j=0}^{3} x_{i,j} - 2\sum_{0 \leq j < l \leq 3} x_{i,j}x_{i,l} + 1$.
- **Column penalties:** For each $j \in \{0,1,2,3\}$, $P_{\text{col}}(j) = \sum_{i=0}^{3} x_{i,j} - 2\sum_{0 \leq i < i' \leq 3} x_{i,j}x_{i',j} + 1$.
- **Diagonal penalties:** There are 7 main diagonals and 7 anti-diagonals. Each contributes pairwise terms with coefficient 1. For example, the main diagonal $k = 0$ contains squares $(0,0), (1,1), (2,2), (3,3)$, yielding $x_{0,0}x_{1,1} + x_{0,0}x_{2,2} + x_{0,0}x_{3,3} + x_{1,1}x_{2,2} + x_{1,1}x_{3,3} + x_{2,2}x_{3,3}$.

The resulting Hamiltonian is $H(x) = \sum_{u \leq v} Q_{u,v}x_u x_v + 8$. The constant offset is $c = 2(4)(1) = 8$.

A feasible solution is given by queens at coordinates $(0,1), (1,3), (2,0), (3,2)$, corresponding to $x_{0,1} = x_{1,3} = x_{2,0} = x_{3,2} = 1$ and all other variables 0. Substituting this assignment into $H(x)$ yields $H(x) = 0$, confirming feasibility and optimality. Any assignment with two queens sharing a row, column, or diagonal will produce $H(x) \geq 1$.